

prob4

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```
[1]: import numpy as np
      from qutip import Qobj, basis, tensor, identity, sigmax, sigmaz

      # Define single-qubit identity and Pauli operators
      I = identity(2)
      X = sigmax()
      Z = sigmaz()
```

```
[2]: # Construct the CNOT using the product-operator expansion
      #  $CNOT_{\{(1 \rightarrow 2)\}} = (1/2) * (II + ZI + IX - ZX)$ 
      CNOT_12 = 0.5 * (tensor(I, I)
                       + tensor(Z, I)
                       + tensor(I, X)
                       - tensor(Z, X))

      # Print or inspect the resulting operator
      print("CNOT gate (control qubit = 1, target qubit = 2):")
      print(CNOT_12)
```

```
CNOT gate (control qubit = 1, target qubit = 2):
Quantum object: dims=[[2, 2], [2, 2]], shape=(4, 4), type='oper', dtype=CSR,
isherm=True
Qobj data =
[[1.  0.  0.  0.]
 [0.  1.  0.  0.]
 [0.  0.  0.  1.]
 [0.  0.  1.  0.]]
```

```
[3]: ket0 = basis(2, 0) # |0>
      ket1 = basis(2, 1) # |1>

      states = {
          "|00>": tensor(ket0, ket0),
          "|01>": tensor(ket0, ket1),
          "|10>": tensor(ket1, ket0),
          "|11>": tensor(ket1, ket1),
      }
```

```

print("Applying CNOT to computational basis states:")
for label, state in states.items():
    out_state = CNOT_12 * state
    print(f"CNOT |{label}> =")
    print(out_state)
    print()

superpos = (states["|00>"] + states["|10>"]).unit()
out_superpos = CNOT_12 * superpos
print("Superposition example:")
print("Initial state: (|00> + |10>) / sqrt(2)")
print("Result after CNOT:")
print(out_superpos)

```

Applying CNOT to computational basis states:

CNOT ||00>> =

Quantum object: dims=[[2, 2], [1, 1]], shape=(4, 1), type='ket', dtype=Dense

Qobj data =

```

[[1.]
 [0.]
 [0.]
 [0.]]

```

CNOT ||01>> =

Quantum object: dims=[[2, 2], [1, 1]], shape=(4, 1), type='ket', dtype=Dense

Qobj data =

```

[[0.]
 [1.]
 [0.]
 [0.]]

```

CNOT ||10>> =

Quantum object: dims=[[2, 2], [1, 1]], shape=(4, 1), type='ket', dtype=Dense

Qobj data =

```

[[0.]
 [0.]
 [0.]
 [1.]]

```

CNOT ||11>> =

Quantum object: dims=[[2, 2], [1, 1]], shape=(4, 1), type='ket', dtype=Dense

Qobj data =

```

[[0.]
 [0.]
 [1.]
 [0.]]

```

Superposition example:

```
Initial state: ( $|00\rangle + |10\rangle$ ) /  $\sqrt{2}$ 
Result after CNOT:
Quantum object: dims=[[2, 2], [1, 1]], shape=(4, 1), type='ket', dtype=Dense
Qobj data =
[[0.70710678]
 [0.         ]
 [0.         ]
 [0.70710678]]
```